H

TEMASEK JUNIOR COLLEGE

2026 JC2 PRELIMINARY EXAMINATION

Higher 2

TEMASEK JUNIOR COLLEGE

CANDIDATE NAME									
CENTRE NUMBER	S					INDEX NUMBER			

PHYSICS **9478/02**

Paper 2 Structured Questions **28 August 2026**

2 hours

You must answer on the Question Paper.

No additional materials are needed

INSTRUCTIONS

- Answer all questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do not use an erasable pen. Do not use correction fluid or tape.
- Do not write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
s.f	
Total	

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\mathrm{m\,F^{-1}})$
elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\mathrm{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$
molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
acceleration of free fall	$g = 9.81\,\mathrm{m\,s^{-2}}$

DO NO

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\mathrm{K} = T/^{\circ}\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

DO NO

Answer **all** the questions in the spaces provided.

1 **(a)** State two differences between *random* and *systematic* errors.

.....

.....

..... [2]

(b) Fig. 1.1 shows a simple pendulum consisting of a mass *m* attached to a light inextensible string of length *L*. The pendulum is secured to a fixed point and made to undergo oscillations when displaced sideways by a small angle *θ*.

DO NOT WRITE IN THIS MARGIN

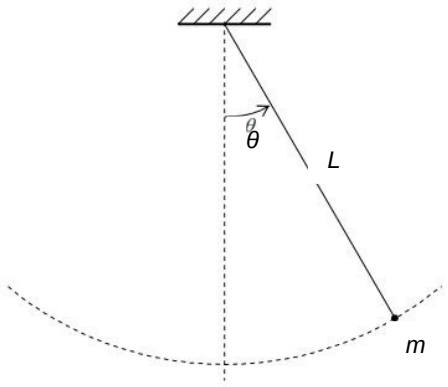


Fig. 1.1 (not to scale)

The period of oscillation *T* can be found by the following equation:

$$T^2 g = 4\pi^2 L$$

where *g* is the acceleration of free fall.

(i) Given $L = 50.0 \pm 0.2$ cm; $g = 9.8 \pm 0.1$ m s^{-2}, find *T* with its associated uncertainty.

T = ... ± ..... s [3]

(ii) When determining period *T* directly using a stopwatch, it is a good practice to take a large number of oscillations.

Explain how this makes the result more reliable.

.....

.....

..... [1]

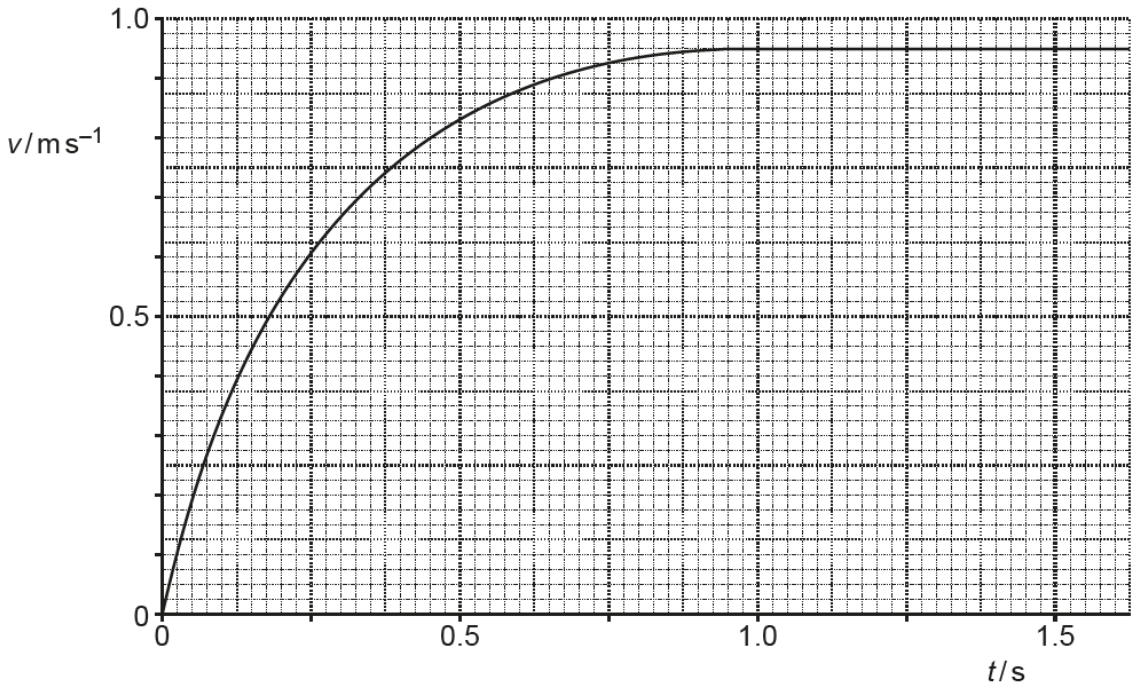
[Total: 6]

2 A student uses a motion-sensor connected to a laptop to investigate the motion of a hollow ball of mass 1.2×10^{-2} kg falling through air.

The ball is dropped from rest. It reaches terminal velocity before it reaches the ground.

The upthrust on the ball is negligible.
Fig. 2.1 shows the variation with time *t* of the velocity *v* of the ball as it falls towards the ground

DO NOT WRI
MAR



DO NO

Fig. 2.1

(a) Using Fig. 2.1, state and explain how the acceleration of the falling object changes over the first 1.5 s.

.....

.....

..... [2]

(b) Use the graph to determine the acceleration of the object at time *t* =

0.25 s after it was released.

acceleration = ... m s^{-2} [2]

(c) Use your answer in **(b)** to determine the drag force acting on the ball at time t = 0.25 s

DO NOT WRITE IN THIS MARGIN

drag force = ... N [2]

DO NO

(d) The student now adds a small amount of sand inside the hollow ball. As before, the ball is dropped from rest and it also reaches terminal velocity before it reaches the ground.

(i) State and explain how the forces acting on the sand-filled ball at v = 0.50 m s^{-1} compares with the forces acting on the hollow ball at this speed.

.....

.....

.....

..... [2]

(ii) State and explain how the terminal velocity of the sand-filled ball compares with the terminal velocity of the hollow ball.

.....

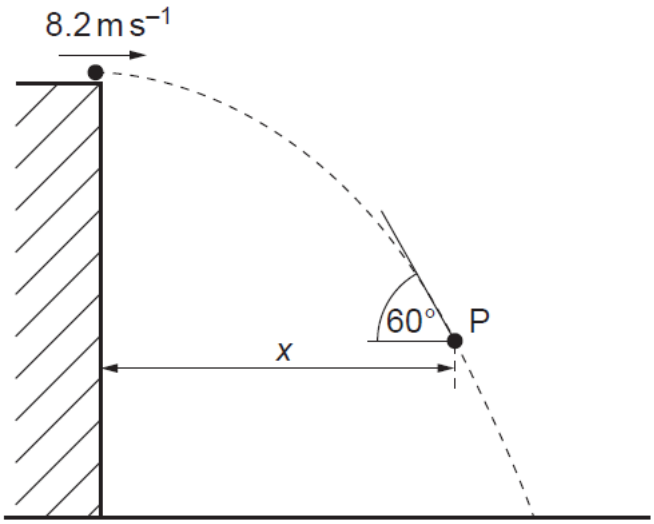
.....

.....

..... [2]

[Total :10]

3 A ball is thrown horizontally from the top of a building, as shown in Fig. 3.1



The ball is thrown with a horizontal speed of 8.2 m s^{-1}. The sides of the building is vertical. At a point P on the path of the ball, the ball is at distance *x* from the building and is moving at an angle of 60° to the horizontal. Air resistance is negligible.

DO NOT WRITE IN THIS MARGIN

DO NO

(a) Explain why the curved path is parabolic.

.....

..... [1]

(b) Calculate

(i) the time through which the ball has fallen from the top to point P

time = ... s [2]

(ii) the horizontal distance *x*

x = ... m [1]

(c) For the movement of the ball from the top to point P, state and explain whether the displacement of the ball is less than, more than or the same as the distance travelled by the ball.

.....

.....

..... [2]

(d) The path of the ball in **(a)**, with an initial horizontal speed of 8.2 m s^{-1}, is again shown in Fig. 3.2.

DO NOT WRITE IN THIS MARGIN

DO NO

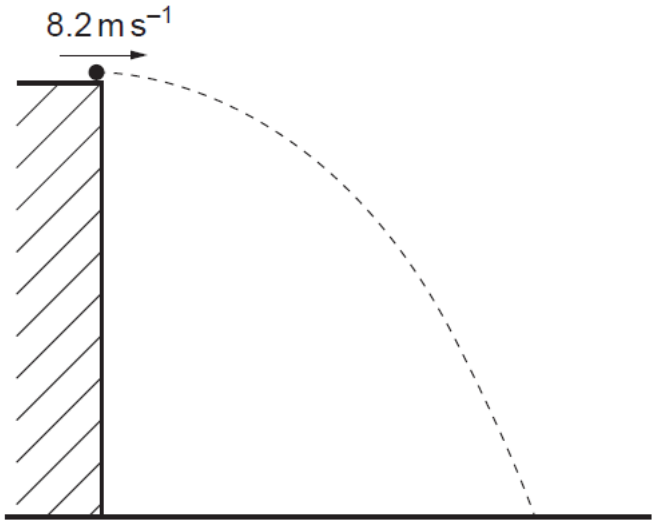


Fig. 3.2

On Fig. 3.2, sketch the new path of the ball having an initial horizontal speed

(i) greater than 8.2 m s^{-1} and with negligible air resistance (label this path G)

(ii) equal to 8.2 m s^{-1} and with air resistance (label this path A)

[3]

[Total : 9]

4 In a binary star system, two stars, each of equal mass 3.5 x 10^{30} kg, rotate about their common

centre of mass O which is equidistant from the centres of the stars, as shown in Fig. 4.1. The separation between the centres of the two stars is 2.0×10^{11} m.

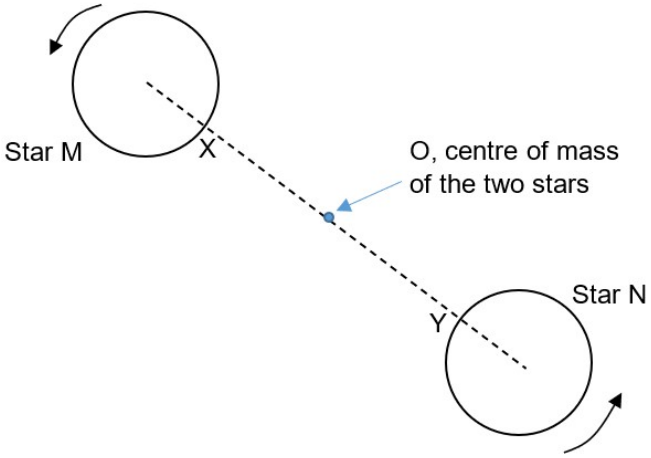


Fig. 4.1

(a) Define gravitational potential at a point.

.....

DO NOT WRITE IN THIS MARGIN

DO NO

.....

.....

[2]

(b) Calculate the gravitational potential at O, the centre of mass of the binary star system.

gravitational potential = ... J kg^{-1} [2]

(c) An asteroid passes through point O, at a speed *v*.

Determine the minimum speed *v* the asteroid must have if it is to escape from the gravitational pull of the binary star system.

minimum speed *v* = ... m s^{-1} [2]

(d) On Fig. 4.2, sketch a graph showing the variation with distance of gravitational potential along the line XY between the two stars. Numerical values are not

required. [2]

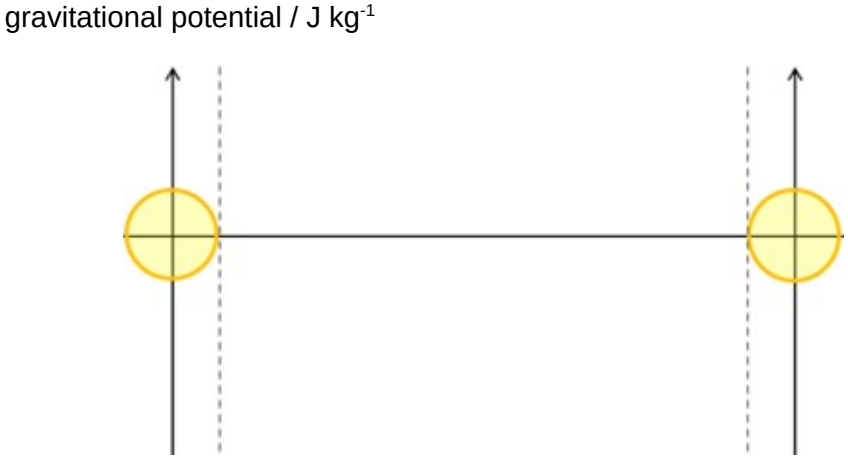


Fig. 4.2

(e) Calculate the time taken for the stars to make one complete revolution about their common centre of mass.

DO NOT WRITE IN THIS MARGIN

DO NO

time = ... s [2]

[Total :10]

5 The first law of thermodynamics can be represented by the expression

$$\Delta U = q + w .$$

(a) State what is meant by the symbols in the expression.

.....

.....

..... [2]

(b) A fixed mass of an ideal gas undergoes a cycle $ABCA$ of changes, as shown in Fig. 5.1.

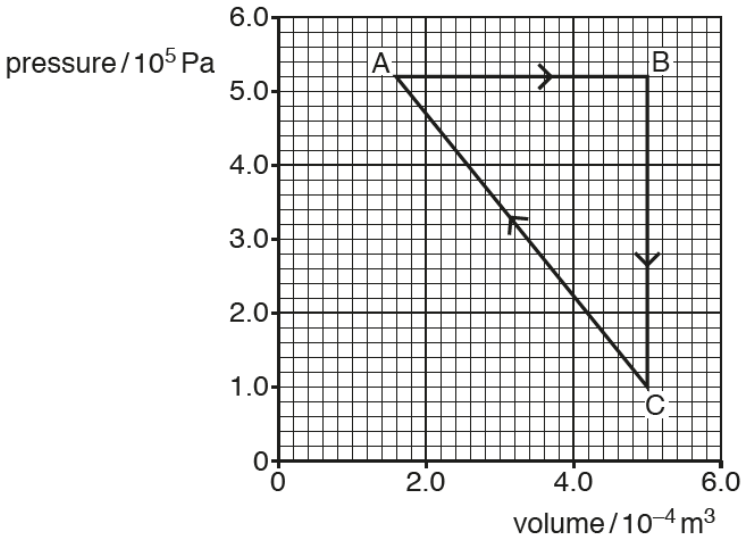


Fig 5.1

DO NOT WRITE IN THIS MARGIN

DO NO

(i) During the change from A to B, the energy supplied to the gas by heating is 442 J. Use the first law of thermodynamics to show that the internal energy of the gas increases by 265 J

[2]

(ii) During the change from B to C, the internal energy of the gas decreases by 313 J. By considering molecular energy, state and explain qualitatively

the change, if any, in the temperature of the gas.

.....

.....

.....

..... [3]

(iii) Calculate the change in internal energy from C to A.

DO NOT WRITE IN THIS MARGIN

change in internal energy = ... J [1]

DO NO

(iv) The temperature of the gas at point A is 227°C.

Calculate the number of molecules in the fixed mass of gas.

Number of molecules = ... [2]

[Total :10]

6 (a) A beam of white light, consisting of wavelengths between 400 nm and 700 nm, is incident on a cool sodium gas at low pressure. Fig. 6.1 shows the four lowest energy levels of the sodium atom.

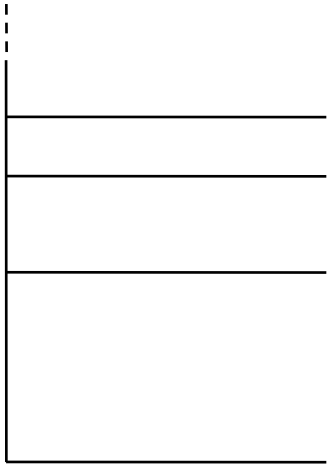


Fig. 6.1

(i) Explain the formation of an absorption line spectrum.

.....

.....

DO NOT WRITE IN THIS MARGIN

DO NO

.....

.....

.....

.....

.....

..... [3]

(ii) Show that the wavelength of the photon that causes the transition from E_1 to E_2 is 592 nm.

[1]

(iii) Determine the number of absorption spectral lines that are observed. Show your working.

number of observable lines = ... [2]

(b) An electron microscope uses an array of electromagnetic lenses to accelerate and focus a beam of electrons to create an enlarged image of a specimen.

(i) Determine the kinetic energy, in eV, of an electron which has a de Broglie wavelength that is the same as the wavelength of the photon in **(a)(ii)**.

DO NOT WRITE IN THIS MARGIN

DO NO

kinetic energy = ... eV [2]

(ii) With reference to the kinetic energy of an electron, explain one advantage of an electron microscope over an optical microscope.

..... [2]

(c) The radial wavefunction of the outermost electron in a sodium atom is approximated by

$$\psi(r) = A\,e^{-r/a},$$

where A and a are constants, and r is the distance from the nucleus.

(i) State the physical meaning of the quantity $|\psi(r)|^2$.

..... [1]

(ii) State and explain where the electron is most likely to be found in the atom.

.....

.....

.....

..... [2]

[Total :13]

DO NOT WRITE IN THIS MARGIN

DO NO

7 In the 1930s, scientists discovered that when a slow neutron is absorbed by a uranium-235 nucleus, the nucleus splits into two smaller nuclei in a process known as nuclear fission. The products have a slightly smaller total mass than the original nucleus, and the missing mass is converted into energy according to Einstein's equation. Each fission reaction also releases, on average, 2.5 neutrons, which can induce further fission reactions. Under

suitable conditions, this leads to a chain reaction. In a nuclear reactor, the chain reaction is carefully controlled to produce a continuous supply of energy. Fig. 7.1 illustrates a typical fission reaction.

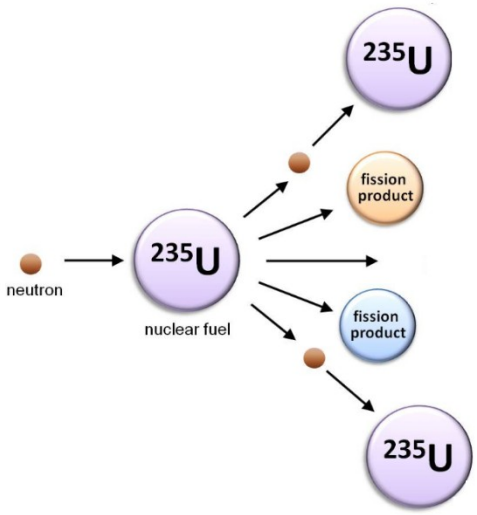


Fig. 7.1

When the conditions are suitable, a chain reaction can occur. If this chain reaction is not controlled, an explosion is likely to occur. However, if the chain reaction is controlled, as in a nuclear reactor, a source of continuous power may be created.

DO NOT WRITE IN THIS MARGIN

DO NO

(a) **(i)** Explain what is meant by neutron-induced fission.

.....

.....

..... [1]

(ii) Explain what is meant by chain reaction.

.....

.....

..... [1]

(iii) Suggest why, in an uncontrolled chain reaction where all neutrons are captured by Uranium-235 nuclei, the majority of the energy is released during the final stages of the fission of a sample of the uranium.

.....

.....

.....

.....

.....

..... [1]

(b) The induced fission reaction of Uranium-235 may be represented by a nuclear equation of the form

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{a}_{b}P + {}^{c}_{d}Q + (2 \text{ or } 3)\,{}^{1}_{0}n + 198\text{MeV}$$

The fission products P and Q have approximately equal masses. However, when any two nuclei are fissioned, the fission products may not be the same. If a large sample of Uranium- 235 is fissioned, many different fission products will be produced. The percentage amount of each fission product in the fissioned material is referred to as the percentage yield.

The variation with nucleon number of the percentage yield of different fission products is referred to as a 'fission yield curve' and is illustrated in Fig. 7.2.

DO NOT WRITE IN MARGIN

DO NO

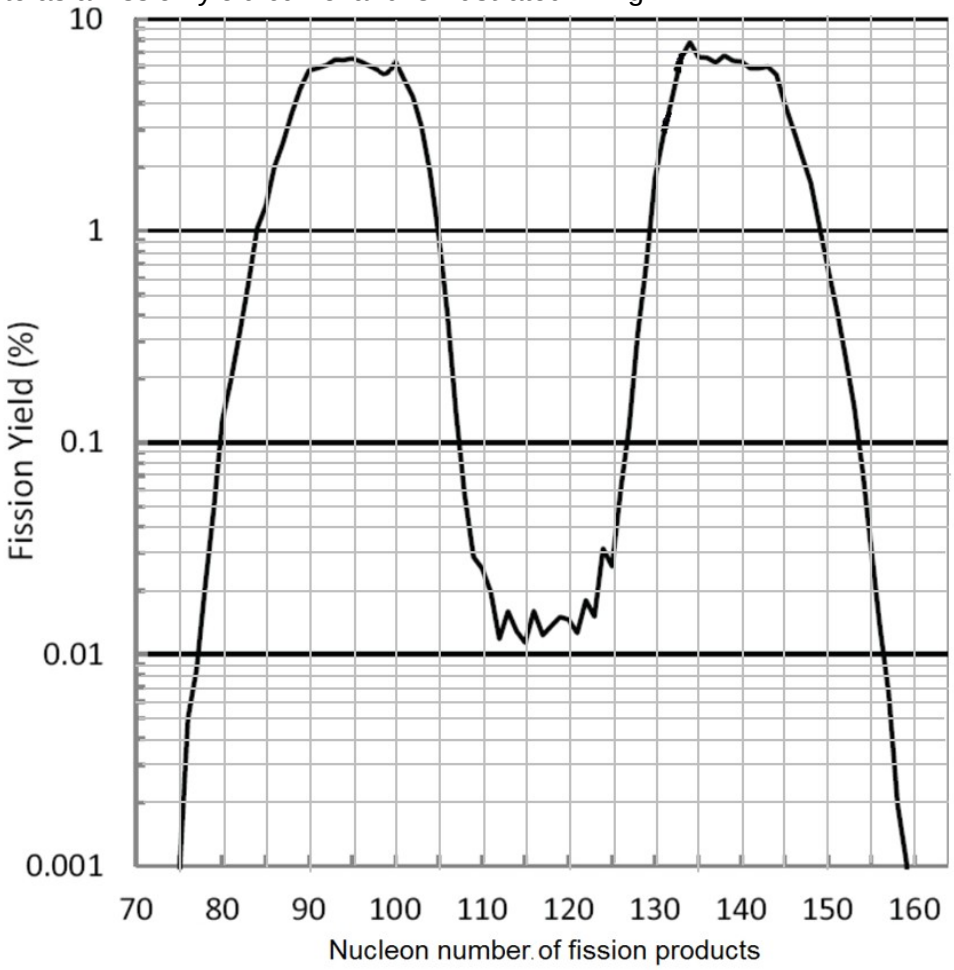


Fig. 7.2

(i) State the proton number of the other fission product for the fission of one uranium if one of the products is $^{89}_{36}Kr$.

proton number = ... [1]

(ii) Suggest why the percentage yield is shown on a logarithmic scale.

.....

.....

..... [1]

(iii) Show that the percentage yield of Molybdenum-99 ($^{99}_{42}Mo$) is about 400 times more than those fission products having approximately equal masses.

[2]

DO NOT WRITE IN THIS MARGIN

(c) The energy released during one fission reaction of a uranium nuclei occurs partly as kinetic energy of the fission products (167 MeV) and of the neutrons (5 MeV).

(i) Suggest one other mechanism by which energy is released in the fission reaction.

..... [1]

(ii) In a nuclear power station, 25% of the energy of the fission products is converted into electrical energy.
Calculate, for the fission of a mass of 1.0 kg of Uranium-235,

1. the number of nuclei in 1.0 kg of Uranium-235,

number of nuclei = ... [2]

2. the electrical energy generated,

energy = ... J [2]

3. the average power output, in megawatts, of the power station if the uranium is fissioned in a time of 24 hours.

power = ... MW [2]

(d) The fission products are usually radioactive and give rise to a series of radioactive decay products. Each decay product has its own half-life. Two such fission products with their decay products and half-lives are shown below.

DO NOT WRITE IN THIS MARGIN

$^{99}_{42}Mo \xrightarrow{67\ hours} {}^{99}_{43}Tc \xrightarrow{2 \times 10^5\ years} {}^{99}_{44}Ru$ (stable solid)

$^{140}_{54}Xe \xrightarrow{16\ seconds} {}^{140}_{55}Cs \xrightarrow{1.1\ minutes} {}^{140}_{56}Ba \xrightarrow{13\ days} {}^{140}_{57}La \xrightarrow{40\ hours} {}^{140}_{58}Ce$ (stable solid)

Consider equal amounts of these two products.
Suggest why there are very different problems for the storage of this nuclear waste.

.....

.....

.....

..... [2]

[Total :16]